Giant Footprints Hint at a Cougar-Size Mammal Among Dinosaurs

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Picture a moonlit night, about 130 million years ago, in what is now southern Brazil. A large mammal pads across damp sand dunes, leaving tracks five inches wide in its wake.

Quarry workers uncovered those **fossilized** footprints in 2019. Scientists still do not know exactly which species made them. But the prints are nearly twice as big as the next-largest mammal tracks from the same **deposits**, and careful study showed the animal walked on all fours, with short legs ending in paws. The conclusion: a roughly cougar-size beast, bigger than any other mammal known from its time.

"Their exceptional size was immediately striking," said Hermínio Ismael de Araújo-Júnior, a **paleontologist** at Rio de Janeiro State University and co-author of a recent study describing the creature in the Journal of South American Earth Sciences.

The find challenges an old idea. Since the 19th century, many scientists assumed mammals stayed small while dinosaurs ruled. The tracks join other clues that some early mammals grew large after all. In Poland, bones of Lisowicia, a pig-like, elephant-size **herbivore** from about 205 million years ago, show an animal over eight feet tall at the shoulder and weighing more than five tons. In eastern China, a badger-like mammal called Repenomamus was found with its teeth sunk into the ribs of a small horned dinosaur about 125 million years ago.

Most mammals of the dinosaur age were still small, often smaller than a housecat, and many had dark coats that likely helped them hide at night. Still, as **paleontologist** Elsa Panciroli of National Museums Scotland put it, the fossil record is incomplete, and more large mammals may be waiting to be found. "As we discover more about **Mesozoic** mammals," she said, "it's clear that they were flourishing."

Word Treasure

fossilized -- Turned to stone over a very long time.
paleontologist -- A scientist who studies fossils and ancient life.
herbivore -- An animal that eats only plants.
Mesozoic -- The age of dinosaurs, long ago in Earth's history.
deposits -- Layers of sand or rock in which fossils are buried.

Background you need

Fossils are the remains or traces of living things from long ago, preserved in rock, and footprints can fossilize when tracks in soft ground are buried and harden over time.

The Mesozoic was the age when dinosaurs lived, and mammals of that time were long thought to be mostly tiny animals that came out at night.

Paleontologists study fossils to learn how ancient animals lived, how big they were and how they moved, even when only bones or footprints survive.

Quarries are places where stone is dug from the ground, and workers there sometimes uncover fossils hidden in the rock.

Break it down

LEAD: Fossil footprints in southern Brazil point to a cougar-size mammal that lived about 130 million years ago.

KEY EVENT: Quarry workers uncovered the fossilized footprints in 2019.

+- WHO: Scientists studying the tracks, including Hermínio Ismael de Araújo-Júnior and Elsa Panciroli.

+- WHAT: Tracks five inches wide, nearly twice as big as the next-largest mammal tracks from the same deposits.

+- HOW: The animal walked on all fours, with short legs ending in paws.

EVIDENCE: Lisowicia bones in Poland and a Repenomamus fossil in eastern China show other early mammals also grew large.

OPEN QUESTION: Scientists still do not know exactly which species made the footprints.

Why it matters

These footprints change what we thought we knew about life in dinosaur times, showing that mammals were not all tiny, and that more surprises may still be buried in the rocks.

Test what you remember

1. How wide were the fossil footprints found in Brazil?

- (A) Two inches wide
- (B) Five inches wide
- (C) One foot wide
- (D) Ten inches wide

2. When were the fossil footprints uncovered?

- (A) In 2005
- (B) In 2019
- (C) In 1990
- (D) In 2024

3. About how big was the animal that made the tracks?

- (A) About the size of a cougar, bigger than any other mammal known from its time
- (B) Smaller than a housecat
- (C) About the size of a small dog
- (D) About the size of an elephant

4. What was Lisowicia, whose bones were found in Poland?

- (A) A small horned dinosaur
- (B) A pig-like, elephant-size herbivore
- (C) A flying reptile
- (D) A badger-like mammal

5. What was found in eastern China?

- (A) A fossilized forest
- (B) A cougar-size mammal's footprints
- (C) A nest of dinosaur eggs
- (D) A badger-like mammal with its teeth sunk into the ribs of a small horned dinosaur

6. What were most mammals of the dinosaur age like?

- (A) Able to fly
- (B) Small, often smaller than a housecat
- (C) As big as cougars
- (D) Bigger than elephants

Answer key (try the quiz first!): 1: B 2: B 3: A 4: B 5: D 6: B

Questions to think about

1. Many scientists since the 19th century: Assumed mammals stayed small while dinosaurs ruled.

2. Paleontologist Elsa Panciroli: Says the fossil record is incomplete and early mammals were flourishing, so more large mammals may be waiting to be found.

MY THOUGHTS (at least 20 words):